Distributed Systems Answers

Q1. Network partitions are unavoidable, so designers must choose between consistency and availability.
Q2. Linearizability guarantees immediate consistency, while eventual consistency allows temporary divergence, such as DNS.
Q3. Vector clocks track causality using logical counters, unlike physical clocks which suffer from clock skew.
Q4. Split-brain happens when multiple leaders are active, causing data corruption.
Q5. W + R > N ensures overlapping reads and writes; without it, stale data may be returned.
Q6. 2PC blocks on failures, while Saga avoids blocking but sacrifices atomicity.
Q7. Raft requires up-to-date logs to ensure leader safety and log consistency.
Q8. CRDTs guarantee convergence; G-Counters merge by taking per-node maximums.
Q9. Circuit breakers have closed, open, and half-open states to stop cascading failures.
Q10. Range partitioning preserves order, hash partitioning balances load; consistent hashing minimizes rebalancing.